Exam 2.E

Practice / Class Review

Name: _____ Section: ◯ 12 ◯ 29

Show all work to earn credit. The exam is worth 102 points; each question is worth 6 points. Scientific calculators only. No graphing calculators, phones, or sharing calculators.

1. Factor completely: $25x^2 - 36y^2$

power of 2 so assume everything is.

$1^2 = 1$
$2^2 = 4$
$3^2 = 9$
$4^2 = 16$
$5^2 = 25$
$6^2 = 36$

$(5x)^2 - (6y)^2$

$\boxed{(5x - 6y)(5x + 6y)}$

2. Solve the inequality, write your answer in interval notation, and graph the solution on the number line.

$-13 < 3x - 1 \leq 5$ ← focus on the one that has an x.

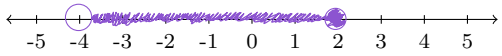
$+1 \quad +1 \quad +1$

$-12 < 3x \leq 6$

$3 \quad 3 \quad 3$

$-4 < x \leq 2$

IN: $(-4, 2]$



3. Solve and write the solution in interval notation: $-9x + 5 < -3(x - 7)$

$-9x + 5 < -3x + 21$

$+3x \quad +3x$

$-6x + 5 < 21$

$-5 \quad -5$

$-6x < 16$

$-6 \quad -6$

multiplied by a neg. so sign switched

$x > \frac{16}{-6}$

x is bigger than 16/-6 so write as

$16/-6 < x$

theres a lower bound but no upper bound, so

$\frac{16}{-6} < x < \infty$

$\boxed{(16/-6, \infty)}$

4. Multiply and simplify completely: $(x + 4)(x^2 - 2x + 3)$

$x^3 - 2x^2 + 3x + 4x^2 - 8x + 12$

$\boxed{x^3 + 2x^2 - 5x + 12}$

5. Solve. Give your answer as an interval: $3|x - 1| + 2 \leq 14$

$-2 \quad -2$

$3|x-1| \leq 12$

$3 \quad 3$

$|x-1| \leq 4$

$x - 1 \leq 4$ and $x - 1 \geq -1$

$+1 \quad +1 \qquad +1 \quad +1$

$x \leq 5 \qquad x \geq 0$

Because and we do it where shared.



$[0, 5]$

6. Factor completely: $2x^3 + 128$

$1^3 = 1$
$2^3 = 8$
$3^3 = 27$
$4^3 = 64$

Do GCD: 2
$2(x^3 + 64)$ ← power of 3, assume everything else is.
$2(x^3 + (4)^3)$
$\boxed{2(x+4)(x^2-4x+4^2)}$

Recall $x^3 + y^3$
$(x+y)(x^2 - xy + y^2)$

7. Factor by grouping: $x^2y + 7y + 4x^2 + 28$

$y(x^2+7) + 4(x^2+7)$
$\boxed{(x^2+7)(y+4)}$

8. Solve the inequality. Write your solution using interval notation: $\dfrac{5x-2}{-4} \geq 7$

$-4\left(\dfrac{5x-2}{-4} \geq 7\right)$ multiply everything and when by a neg. switch signs

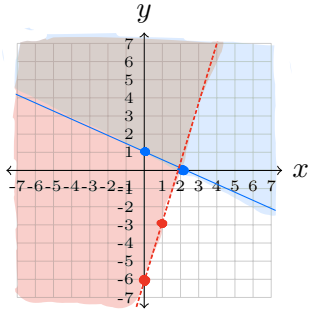
$\dfrac{(-4)(5x-2)}{(-4)} \leq 7(-4)$

$5x - 2 \leq -28$
$+2 \quad +2$
$\dfrac{5x}{5} \leq \dfrac{-26}{5}$
$-\infty < x \leq \frac{-26}{5}$
$\boxed{(-\infty, -\frac{26}{5}]}$

9. Graph the following system: $\begin{cases} x + 2y \geq 2, \\ 3x - y < 6. \end{cases}$

$x + 2y \geq 2$
$-x \quad -x$
$2y \geq -x + 2$
$\dfrac{2y}{2} \geq \dfrac{-x}{2} + \dfrac{2}{2}$
$y \geq -\frac{1}{2}x + 1$
Line is solid
Shade on top of line

$3x - y < 6$
$-3x \quad -3x$
$-y < -3x + 6$
$\dfrac{-y}{-1} < \dfrac{-3x}{-1} + \dfrac{6}{-1}$
$y > 3x - 6$
Line is dotted
shade on top of line



10. Solve the compound inequality, write your answer in interval notation, and graph the solution on the number line.

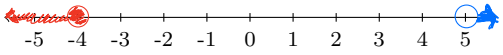
$3x + 7 \leq x - 1$ OR $5x - 6 > 2x + 9$

$-x \quad -x$
$2x + 7 \leq -1$
$-7 \quad -7$
$\dfrac{2x}{2} \leq \dfrac{-8}{2}$
$x \leq -4$

or

$-2x \quad -2x$
$3x - 6 > 9$
$+6 \quad +6$
$\dfrac{3x}{3} > \dfrac{15}{3}$
$x > 5$

$(-\infty, -5] \cup (5, \infty)$



Remember
OR = ∪
AND = ∩

11. Solve the equation: $p(p+3) = 54$

$p^2 + 3p = 54$
$-54 \quad -54$
$p^2 + 3p - 54 = 0$

this sign gives the sign of the bigger factor
sign tells us if both signs are same

54
$-1 + 54 = 53$
$-2 + 27 = 25$
$-3 + 18 = 15$
$-6 + 9 = 3$

$p^2 - 6p + 9p - 54 = 0$
$p(p-6) + 9(p-6) = 0$
$(p-6)(p+9) = 0$

$p - 6 = 0$
$+6 \quad +6$
$\boxed{p = 6}$

$p + 9 = 0$
$-9 \quad -9$
$\boxed{p = -9}$

12. Solve the equation: $2|3x - 5| + 7 = 25$

$-7 \quad -7$

$2|3x-5| = 18$

$\frac{2}{2} \quad \frac{2}{2}$

$|3x-5| = 9$

$3x-5=9$ or $3x-5=-9$

$+5 \quad +5 \qquad +5 \quad +5$

$\frac{3x}{3} = \frac{14}{3} \qquad \frac{3x}{3} = \frac{-4}{3}$

$x = 14/3 \qquad x = -4/3$

13. Factor completely: $18x^2 - 39x + 18$

GCD: 3

$3(6x^2 - 13x + 6)$

$3(6x^2 - 4x - 9x + 6)$

$2(2x(3x-2) - 3(3x-2))$

$2(3x-2)(2x-3)$

$6 \cdot 6 = 36$

$-1 \quad -36 = -37$

$-2 \quad -18 = -20$

$-3 \quad -12 = -15$

$-4 \quad -9 = -13$

$-6 \quad -6 = -12$

14. Factor the polynomial completely: $3t^6 - 6t^5 - 45t^4$

GCD: $3t^4$

$3t^4(t^2 - 2t - 15)$

$3t^4(t^2 + 3t - 5t - 15)$

$3t^4(t(t+3) - 5(t+3))$

$3t^4(t+3)(t-5)$

15

$+1 \quad -15 = -14$

$+3 \quad -5 = -2$

15. Subtract the polynomials and simplify the result completely:

$(7x^4 - 3x^2 + 4x - 9) - (2x^4 + 5x^3 - 8x^2 - 6)$

$7x^4 - 3x^2 + 4x - 9 - 2x^4 - 5x^3 + 8x^2 + 6$

$5x^4 - 5x^3 + 5x^2 + 4x - 3$

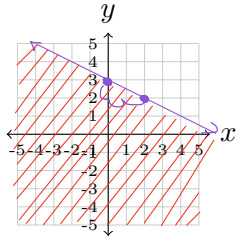
16. Graph: $y \leq -\frac{1}{2}x + 3$

line under means solid line.

y is less than the line so shade under.

(0,3)

$\frac{-1}{2}$



17. The product of two consecutive page numbers is 306. Find the page numbers.

Consecutive number the next number

ex: 4 the next consecutive # is 5

$6 \to 7$

$8 \to 9$

$x \to x+1$

↳ you don't know the number so it x.

$x(x+1) = 306$

$x^2 + x = 306$

$-306 \quad -306$

$x^2 + x - 306 = 0$

$x^2 - 17x + 18x - 306 = 0$

$x(x-17) + 18(x-17) = 0$

$(x+18)(x-17) = 0$

$x+18=0$ or $x-17=0$

$-18 \quad -18 \qquad +17 \quad +17$

$x = -18 \qquad x = 17$

306

$-1 \quad +306 = 305$

$-2 \quad +153 = 151$

$-3 \quad +102 = 99$

$-6 \quad +51 = 45$

$-17 \quad +18 = 1$

But $x = -18$ does not make sense (extraneous solution) the only answer is

$x = 17$